GenderStego: A Novel Framework for Gait-Based Gender Recognition and Video Steganography

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ABSTRACT

GenderStego: A Novel Framework for Gait-Based Gender Recognition and Video Steganography

by Sohan Vallapureddy

The convenience of surveillance technology has raised several serious ethical questions regarding an individual's right to privacy and the security of their biometric information. To address the conflict between the use of surveillance technology to protect society and the need to preserve individual anonymity, we propose a new method called **GenderStego**. GenderStego is a novel framework for gait-based gender recognition in conjunction with privacy-preserving video steganography. Traditional gender recognition methods rely on identifying facial or other visual characteristics. In contrast, this research demonstrates how our system uses a privacy-preserving pipeline to identify a person's gender at long distances using only skeletal motion.

Our proposed system employs a Spatial-Temporal Graph Convolutional Network (ST-GCN) to analyse poses derived from skeletal keypoints tracked using the state-of-the-art keypoint tracker YOLOv8x-Pose, thereby removing the need for visual identification. Our experiments demonstrate a validation accuracy of **72.65%** on the large-scale, in-the-wild DenseGait dataset, without signs of overfitting. We also observe strong evidence that our model continues to improve even after reaching the maximum training budget of 20 epochs.

We additionally incorporate a privacy module that removes the subject's visible appearance using YOLOv8-based object segmentation and HSV Gaussian inpainting, overlaying the subject with a blank background and colour-coded skeletons (blue: Male; red: Female), thereby providing comprehensive protection of biometric information contained in the original frames. The original video is then embedded into this skeletal output using a modified Discrete Cosine Transform (DCT) steganographic

scheme applied to 4×4 frequency-domain blocks. For added security, we encrypt the hidden payload with AES-256-CTR for confidentiality and append an HMAC-SHA256 signature for tamper detection, enabling authorised entities to recover the data for forensic analysis while keeping sensitive biometric information private from unauthorised parties.

In addition to qualitative evaluations across multiple parameter settings, our results indicate that the recommended configuration ($\mathrm{QP} = 27$, $\beta = 30$, $\alpha = 0.05$) yields a stego PSNR of 34.48 dB—below the threshold of human perceptual detection—and a reconstruction PSNR of 36.51 dB with 99.34% cosine similarity relative to the original video. Alongside quantitative testing on six real-world surveillance videos, which demonstrates stable behaviour under typical conditions and graceful degradation in challenging environments, GenderStego emerges as one of the first systems to combine robust security with accurate video analytics for ethical AI and responsible surveillance.

Keywords: GenderStego, Gait-based Gender Recognition, ST-GCN, Video Steganography, DCT, AES-256-CTR, HMAC, Biometric Privacy, Ethical AI, Computer Vision, YOLOv8

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CHAPTER 1

INTRODUCTION

Computers now recognize, classify, and analyze human behavior much more easily thanks to the integration of computers into everyday life and the easy-to-use nature of today's technologies. In addition, because they are trained to recognize human behavior and demographics from video, current state-of-the-art computer vision models can easily track individuals in ordinary videos. Even though computer vision has been instrumental in providing us an insight into behavior and allowing us to monitor people for their own safety, the technology also raises serious privacy concerns. Ordinary video recordings can be easily used to reveal the identities of individuals and their private information, without either consent or knowledge. Furthermore, facial features, clothing patterns, body shape, and even object shapes found in the background of the video can be used for both profiling and recognizing individuals --- no matter how far away the individual may be.

All types of behavior and human activities can be tracked by means of tracking how individuals move (their ''gait''), which is particularly important. Gait is a biometric trait. An individual's walking style can be tracked and captured from a distance and without the subject's knowledge or consent. Furthermore, the pattern of one's gait provides information about many aspects of an individual including their gender, age, emotional state, medical or injury condition(s), body weight, height, etc. This is where ethics regarding the collection of technological information come into play. There are clear advantages to using gait for identification; however, there are significant risks involved in revealing personal information and/or an individual's physical identity.

Therefore, this framework addresses obtaining and keeping gait feature information for analysis while removing all identifiable information from the gait data.

Analysis of gait occurs without using the original frame images of the individual. A skeletal keypoint position extraction process is used instead for further analysis. With a masking technique combined with background modeling and inpainting techniques, visual identity is completely eliminated. However, a colour-coded skeletal representation will remain visible. The framework includes embedding the entire original video in the skeletal output through encrypted steganography so that the original video can be stored safely for later retrieval by authorized users.

1.1 Objective

The primary goal of the proposed research was to demonstrate that it is feasible to perform meaningful analysis while protecting sensitive information.

1.2 Privacy Concerns Regarding Video-Dependent Gender Classification

Video-dependent systems may unintentionally reveal:

- facial features,
- clothing, appearance, and skin tones,
- environment and background,
- body structure and behavior patterns.

With increased surveillance and continuous recording, these risks are exacerbated. Unintentional identity recognition, profiling, and demographic inference may occur without either the knowledge or consent of those being recorded. Increasingly, emerging regulatory frameworks like GDPR emphasize minimizing exposure of identifiable data when conducting visually aware processing of data.

Although gender classification based on gait offers substantial value toward understanding crowds, demographics, and public safety applications, the challenge exists in accomplishing this type of analysis without compromising the privacy of individuals in the video.

1.3 Research Goals

The purpose of this research project is to:

- manage privacy concerns associated with video-based gender classification,
- designate a gait-based gender classification model using pose keypoints,
- eliminate all appearance-based identifiers from the video,
- implement a privacy-friendly video processing method (skeletonization),
- securely embed the original video within the skeletonized version using encryption and steganography.

The focus of this project is on developing a system capable of handling video content responsibly while still permitting analytical tasks to produce valid results.

1.4 Relevance

This project is relevant for several reasons:

- surveillance has grown extensively with cameras present almost anywhere we go,
- raw video frames offer potential for disclosure of individual identity,
- demographic or behavioral information may potentially be misused if profiled,
- there are legal ramifications for failing to protect data during processing,
- ethical guidelines support responsible analytics.

GenderStego allows beneficial insights without compromising individual privacy, supporting ethically driven video analytics.

1.5 Goals and Objectives

The major goals and objectives for this project were to:

- develop a gait-based gender classification system using ST-GCN and pose estimation,
- construct a privacy-friendly video processing flow that eliminates identifiable attributes,

- develop methods to store and embed video securely using steganography and encryption,
- generate a skeletal video that preserves movement and gender cues but does not preserve individual identity,
- demonstrate a fully functional system that maintains a balance among privacy, usefulness, and security.

1.6 Report Organization

Chapter 2 contains a detailed analysis of the related literature, focusing on the evolution of gender classification using gait analysis, privacy-preserving strategies, and media encryption. Chapter 3 describes the datasets used and the preprocessing pipeline. Chapter 4 presents the ST-GCN architecture used for gender classification. Chapter 5 provides a detailed walkthrough of each stage of the implementation. Chapter 6 presents quantitative and qualitative experimental results. Chapter 7 concludes the work, and Chapter 8 outlines directions for future development.

CHAPTER 2

LITERATURE REVIEW

Gait has long been recognised as a useful behavioural biometric because it captures a person's natural walking pattern at a distance without requiring contact or high-resolution imagery. Traditional silhouette-based methods are sensitive to lighting and clothing, so recent work has shifted towards pose-based representations that operate directly on joint coordinates. An and colleagues [3] introduced the OU-ISIR Multi-View Large Population database with pose sequences and showed that model-based gait analysis can scale to more than ten thousand subjects across multiple viewpoints. Follow-on systems such as GenReGait [4] and the lightweight gender classifier in [8] demonstrate that keypoint-based deep models can achieve robust gender recognition using joint-level gait features, often with Spatial–Temporal Graph Convolutional Networks (ST-GCNs) for efficient processing on skeletal graphs. At the same time, the broader gait recognition community has explored more expressive sequence models. Ma et al. [5] learn visual prompts with transformer-based attention to improve cross-view recognition, while recent ECCV work on occluded gait [6] uses mixtures of temporal and action experts to handle partial observations. Another ECCV study [7] proposes relation descriptors to capture inter-joint relationships for free performance gains without extra labels. These methods collectively show that pose-based and graph-based architectures are effective for gait and gender classification, but they typically assume access to raw video frames and do not integrate any explicit privacy preservation or downstream secure storage of the original content. WildGait [2] pushes this line of work into raw surveillance streams by pre-training an ST-GCN on automatically annotated walking skeletons from CCTV-like footage, using a large ‘‘Uncooperative Wild Gait’’ dataset of anonymised 2D poses.

In parallel, a separate line of research focuses on privacy-preserving video process-

ing. Elhadad et al. [1] describe an early steganographic approach for hiding private video data in surveillance systems, highlighting the tension between continuous monitoring and subject anonymity. More generally, privacy-aware video pipelines often rely on background modelling and masking to remove identifiable appearance while retaining scene context. HSV Gaussian background models [1] and related techniques build a clean reference frame and then replace human regions using Gaussian statistics in colour space to anonymise identity. Other works apply detector-driven appearance abstraction, for example using YOLO or Mask R-CNN to localise people and then blur, pixelate, or mask faces and clothing rather than reconstructing an entirely person-free scene. Context-aware systems such as X-Stream (noted in the broader privacy literature rather than a single canonical paper) introduce adaptive privacy operators that change the strength of anonymisation based on content and inferred risk. Additional strands explore encrypted-domain analytics, where portions of H.264/HEVC streams are processed without decryption, and differential-privacy style noise injection to limit what can be inferred about any individual from model outputs. These approaches successfully hide visual identity for many surveillance tasks, but they are usually decoupled from learning-based analytics such as gait-based gender recognition and rarely provide a secure, reversible representation of the original video.

Video steganography work, which superficially resembles privacy-preserving pipelines, is mainly concerned with hiding arbitrary payloads imperceptibly rather than with structured analytics. Recent deep steganography systems such as StegaNeRV [9] use implicit neural representations and U-Net style architectures to embed entire videos at high capacity while maintaining visual fidelity. Stego-STFAN [10] introduces a spatial–temporal filter adaptive network that leverages attention and learned filters to preserve temporal consistency in the stego sequence. Other domains have adapted deep video steganography to medical data: for example, the medical

steganography approach in [11] combines a deep CNN with AES-256 encryption to protect sensitive clinical footage. Classical schemes have also been extended, such as octree-based spatiotemporal redundancy exploitation for efficient payload placement [12] or hybrid algorithms that pair AES with edge-aware embedding to balance security and imperceptibility [13]. These methods demonstrate that it is possible to hide large, encrypted payloads inside video streams without obvious artefacts, but they generally operate on full RGB frames and are not explicitly tied to pose-only, privacy-preserved representations. Moreover, they typically treat the hidden video as an opaque payload rather than one that is algorithmically linked to a privacy-aware analytic pipeline.

Taken together, the existing literature reveals three largely separate threads: pose-based gait and gender recognition with strong accuracy but full access to raw video; privacy-preserving video processing that hides appearance but does not support structured demographic analytics; and video steganography that securely stores or transmits original videos but is rarely integrated with skeletal representations or privacy-by-design workflows. To our knowledge there is no prior framework that combines gait-based analysis, identity-safe skeletal video generation, and encrypted embedding of the original footage into a single, end-to-end system. GenderStego is designed to fill this gap by integrating all three components—pose-only gender classification, privacy-preserving visualisation, and secure steganographic storage—into one coherent pipeline.

CHAPTER 3

Dataset and Data Processing

Two separate datasets were utilised in this research. A principal dataset, *DenseGait*, was used for training and evaluating the proposed model. The OU-ISIR Gait Database was initially employed to validate the operation of the ST-GCN architecture and training loop prior to using DenseGait for training.

3.1 DenseGait Dataset

DenseGait is a gait database created by collecting a large number of skeleton sequences from real-world surveillance video. Skeletons are identified by automated object tracking in video files and are annotated using an automated attribute detection process. The dataset contains approximately 217,000 labelled skeletons with 42 gait-related attributes. One of these attributes, sex, is a binary label used as the classification target in this research.

A variety of realistic surveillance conditions were incorporated into the development of DenseGait, including partial occlusion, variable lighting, multiple camera viewpoints, and varying subject–camera distances. Owing to these conditions, DenseGait presents significantly more challenging data than laboratory-controlled databases and more closely reflects the operational conditions of real-world surveillance systems as illustrated in Figure 1.



Figure 1: Example pose sequences from the DenseGait surveillance dataset [3]

3.1.1 Rationale for Using DenseGait in GenderStego Model Training

DenseGait was selected as the main training dataset for the *GenderStego* model for three primary reasons. First, because DenseGait was collected in diverse real-world environments, models trained on this dataset are expected to generalise beyond laboratory-controlled settings. Second, with over 217,000 tracklets, the dataset is large enough to support deep learning without requiring manual labelling of tracklets. Third, since the binary gender label is directly provided as one of the 42 automatically determined attributes in DenseGait, no additional labelling effort is required for gender classification.

3.1.2 Input Format and Label Conventions

Each skeleton sequence is represented as a tensor of shape $[\text{Batch}, 3, T, 18, 1]$. The three channels correspond to the x-coordinates, y-coordinates, and confidence scores of each of the 18 body joints. The temporal dimension T is standardised to 64 frames by uniformly subsampling longer sequences or zero-padding shorter ones. The gender label follows the convention $\text{Male} = 0$ and $\text{Female} = 1$.

3.1.3 Limitations of DenseGait

Because the gender labels in DenseGait are generated automatically by attribute detection algorithms rather than manually annotated by human experts, some degree of label noise is likely, which may impose a practical limit on the maximum achievable performance of the model. Furthermore, DenseGait includes only binary representations of gender, so no conclusions can be drawn about the model's ability to identify individuals outside the male/female categories. Demographic metadata beyond the 42 automatically detected attributes is not available; thus, fairness evaluations across other demographic dimensions (e.g. age, ethnicity, or body type) cannot be performed using DenseGait alone.

3.2 OU-ISIR Gait Database

The OU-ISIR Gait Database with Pose Sequences is a multiview gait dataset designed specifically for gait analysis. It contains 10,307 subjects who perform a round-trip walking sequence. Ten synchronised cameras are placed at 15° intervals around the walking path, yielding 14 distinct viewpoints when walking both directions. All images are captured at a resolution of 1280 × 980 pixels and 25 frames per second.

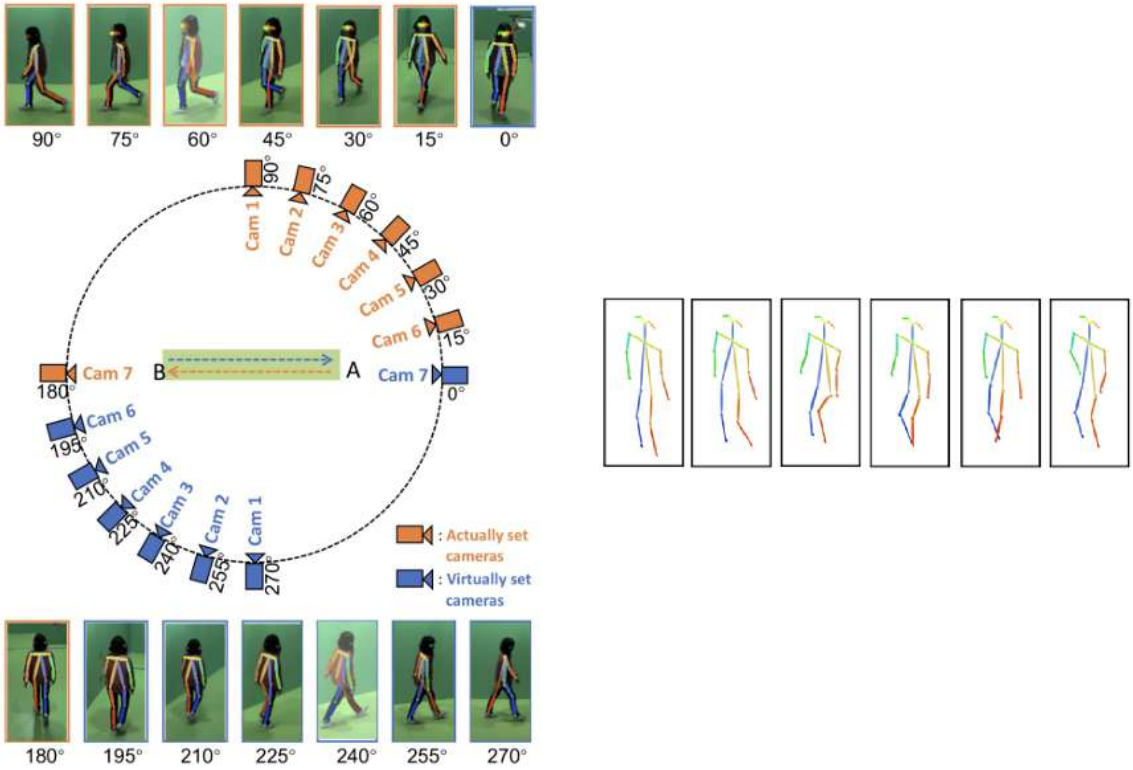


Figure 2: Camera layout and example multi-view pose sequences from the OU-ISIR multi-view gait dataset, adapted from An et al. [3].

Figure 2 contrasts the controlled, multi-view OU-ISIR setting with the noisier, in-the-wild DenseGait sequences.

3.2.1 Pose Extraction Methods

Pose sequences are extracted from RGB walking videos using two modern pose estimation frameworks: OpenPose (Cao et al., CVPR 2017) and AlphaPose (Fang

et al., ICCV 2017). These methods detect 18 major body joints in each frame of the walking video. In addition to the (x, y) coordinates for each joint, they also provide a confidence score per joint. Typical sequences range from 18 to 35 frames per walk cycle.

3.2.2 Pose Sequence Representation

Each pose sequence consists of a series of frames covering a full walk cycle. Each frame encodes the (x, y) coordinates and confidence scores for each of the 18 joints, yielding a tensor of shape (frames $\times$ 18 $\times$ 3). This representation enables the construction of both spatial and temporal graph structures. Spatial graphs model anatomical relationships among joints within each frame, while temporal graphs capture how joint positions evolve between consecutive frames. Together, these structures support modelling of gait dynamics using spatial–temporal graph-based models.

3.2.3 Preprocessing Steps for ST-GCN Model Input

All input pose sequences, regardless of whether they are obtained from a single video stream or from multiple camera streams (i.e. multi-camera), undergo a consistent data preparation process prior to being used to train the proposed ST-GCN-based model. The following steps are always applied:

1. **Frame count normalisation:** The frame count of each individual sequence is normalised to a fixed length of 64 frames. Thus, irrespective of the original sequence length, every sequence fed into the model contains exactly 64 frames.

2. **Joint coordinate normalisation:** Joint coordinate values are normalised so that positional information becomes independent of subject height and camera position. Joints are defined based on their anatomical locations, ensuring that relative joint positions are preserved even when subjects differ in height or are

captured from different viewpoints.

3. **Format conversion:** The 17 COCO-format joints identified by the YOLOv8x-Pose model are converted to the 18-joint OpenPose format. An additional Mid-Hip joint is created at the midpoint between the left and right hip joints, and one of the original joints is replaced by this new joint.

4. **Handling missing or low-confidence joints:** Missing detections and low-confidence joint scores are either smoothed or replaced with zeros to maintain temporal continuity across frames.

5. **Skeleton graph construction:** Anatomically defined intra-frame and inter-frame links between joints are used to establish both spatial and temporal relationships. These connections provide the fixed graph structure required to apply spatial–temporal graph convolutions.

6. **Gender label assignment:** Each pose sequence is assigned a binary gender label based on the metadata associated with the sequence, following the convention Male $= 0$ and Female $= 1$.

Through this pipeline, all pose sequences are standardised, formatted, and cleaned so that they can be effectively used as inputs to the ST-GCN model.

CHAPTER 4

Overview of ST-GCN for Gender Classification

4.1 Pose Estimation Process

Gender classification in GenderStego is performed using a Spatial--Temporal Graph Convolutional Network (ST-GCN), a model specifically designed to analyze human motion from pose sequences. Unlike appearance-based gender recognition methods that rely on facial features or clothing, ST-GCN operates entirely on skeletal joint data extracted from video frames. This ensures that the classification process remains privacy-preserving from the very beginning.

ST-GCN is particularly suited for gait analysis because it processes both the spatial structure of body joints and their temporal evolution during walking. Gait contains subtle cues---stride rhythm, limb coordination, hip rotation, and pelvis sway--many of which carry gender-distinctive information. By learning these temporal and structural patterns, ST-GCN can recognise gender reliably without accessing any identity-sensitive visual detail.

In the GenderStego framework, ST-GCN receives pose sequences as input and outputs a predicted gender label. This prediction is later used for skeleton colour-coding in the privacy-preserving visualisation stage. The model is trained on the DenseGait dataset, which provides large-scale, in-the-wild skeleton sequences with binary gender labels, allowing the classifier to learn consistent gender-related gait characteristics across a wide range of real-world conditions.

4.2 Graph Representation of the Human Skeleton

To analyze gait using ST-GCN, the human skeleton is translated into a spatio-temporal graph structure that forms the core input representation. Each joint in the

skeleton is treated as a node. Within a single frame, joints are connected by spatial edges corresponding to anatomical links---for example, connections between the hip and knee, knee and ankle, or shoulder and elbow. These edges preserve the geometric structure of the human body and encode posture information.

To capture motion, temporal edges connect each joint in one frame to the same joint in the next frame. This creates a chain of joint movements across time, enabling the model to observe changes in posture, rhythm, and body coordination. Walking is inherently a temporal activity, and these sequential dependencies are essential for identifying gender-specific gait cues such as stride length and hip-sway amplitude.

Because the ST-GCN operates solely on this structured skeletal graph, it naturally avoids visual details such as colour, texture, or clothing. This ensures that the gender classification model is fully privacy-preserving, relying strictly on motion dynamics encoded within the pose sequences.

4.3 Gender Classification Workflow

The gender classification workflow follows a clear sequence:

- Skeleton sequences and gender labels are loaded from the DenseGait dataset, which provides 18-joint coordinates for each tracklet along with a binary gender attribute.
- The pose data is converted into a spatio-temporal graph, where joints and their connections are arranged according to predefined spatial and temporal relationships.
- The graph is passed through three progressive ST-GCN stages, expanding channel depth from 64 to 128 to 256, which allow the model to learn both short-range posture cues and long-range temporal gait patterns.

- After the final stage, global average pooling collapses the temporal and joint dimensions into a single feature vector.
- A fully connected layer and sigmoid activation produce a binary gender score. Scores below 0.5 are classified as Male and scores of 0.5 or above as Female.

Because the model is trained on thousands of in-the-wild tracklets across diverse lighting and camera conditions, it becomes robust to variations in movement style, viewpoint, and body proportions. The ST-GCN module enables demographic analysis while ensuring that the system never relies on raw video appearance---only on motion encoded through pose estimation.

CHAPTER 5

COMPONENT ANALYSIS AND IMPLEMENTATION OF STAGES

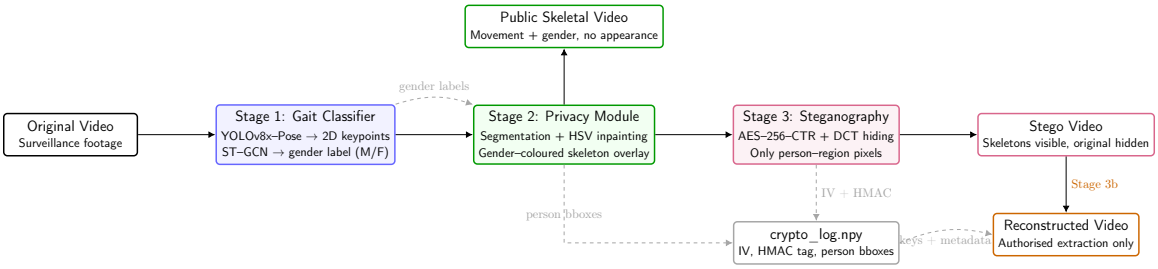


Figure 3: Overall architecture of GenderStego.

Chapter 5 presents a detailed walk-through of the implementation of each of the three components in the GenderStego pipeline: Stage 1 (gait-based pose estimation and gender classification), Stage 2 (privacy-preserving video processing), and Stage 3 (secure steganographic embedding and reconstruction).

5.1 Stage 1: Pose Estimation and Gender Classification

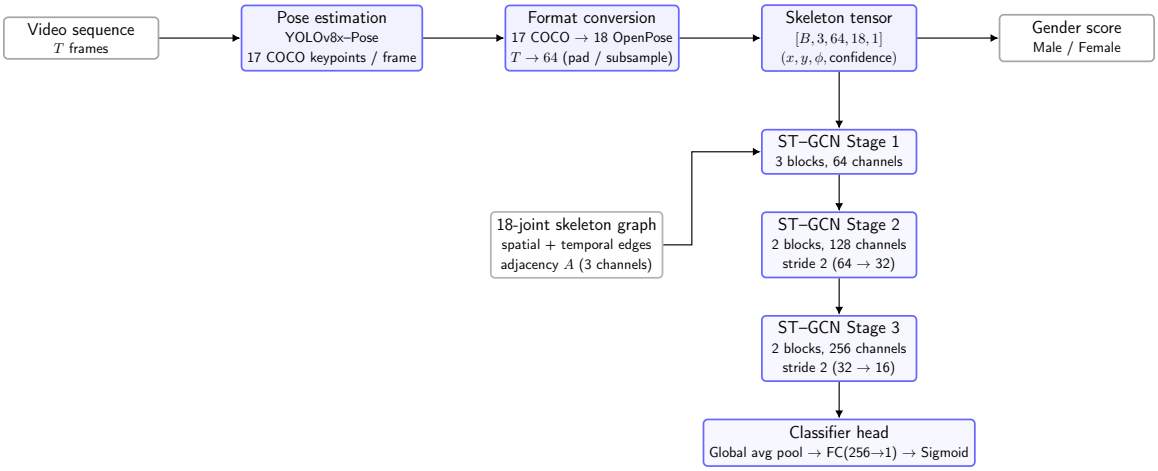


Figure 4: Stage 1: gait-based pose estimation and gender classification using YOLOv8x-Pose and ST-GCN.

5.1.1 Pose Estimation

The first component of the GenderStego pipeline involves extracting skeleton poses from all frames in the input video. The final implementation uses YOLOv8x-

Pose for keypoint detection. YOLOv8x-Pose is a one-stage architecture that detects individuals within a frame and concurrently estimates the locations of 17 body joints using the COCO keypoint representation, along with a confidence score for each joint.

Earlier versions of the GenderStego pipeline used an OpenCV DNN-based wrapper for OpenPose for keypoint detection. This approach was abandoned in favour of YOLOv8x-Pose because the latter provides faster inference and improved detection performance under realistic conditions such as partial occlusion and varying illumination.

To ensure consistent tracking across frames, a centroid-based person tracking algorithm is applied following detection. The Euclidean distance between newly detected centroids and centroids from the immediately preceding frame is computed. If this distance is less than 200 pixels, the detection is associated with an existing active tracking ID; otherwise, a new tracking ID is created. Tracking IDs start at zero and are incremented by one for each new track. Consequently, even if a person temporarily leaves the field of view, multiple IDs are never assigned to the same individual.

Since the DenseGait training data uses the 18-joint OpenPose format, a conversion step maps the 17 COCO joints to the 18 OpenPose joint locations before applying the gender classification model. An additional mid-hip joint is synthetically derived as the midpoint between the left and right hip joints. This joint encodes information relevant for gender classification, as differences between men and women in hip sway and pelvis width can be observed in gait.

All skeleton sequences are normalised to a maximum length of 64 frames. Sequences shorter than 64 frames are zero-padded at the end, while longer sequences are uniformly sub-sampled over time. The resulting input tensor has shape $[B, 3, 64, 18, 1]$, where B denotes the batch size, the three channels encode the x-coordinate, y-

coordinate, and confidence value for each of the 18 joints, and the final dimension indicates that there is a single person per sample.

5.1.2 Gender Classification Using GaitSTGCN

The gender classification model is a spatial–temporal graph convolutional neural network (ST-GCN) that extracts gender-relevant features from pose sequences independently of visual appearance. The model, referred to as *GaitSTGCN*, is organised into three successive phases that progressively increase the channel depth while reducing temporal resolution.

Phase I consists of three consecutive ST-GCN blocks operating at 64 channels over the full 64-frame temporal sequence. Phase II contains two blocks that increase the channel depth to 128 while reducing the temporal length to 32 frames. Phase III includes two further blocks that increase the channel depth to 256 and reduce the temporal length to 16 frames.

After Phase III, global average pooling is applied over both the temporal and joint dimensions to produce a 256-dimensional feature vector. This vector is passed through a fully connected layer followed by a sigmoid activation to yield a binary gender score in the range $[0, 1]$. Scores below 0.5 are classified as Male, while scores greater than or equal to 0.5 are classified as Female.

Each ST-GCN block comprises a spatial graph convolution, a temporal convolution with kernel size 9, and a ReLU activation. The spatial graph convolution operates over a fixed adjacency matrix that encodes the 18-joint OpenPose skeleton topology. At each spatial convolution step, learnable edge-importance weights are multiplied element-wise with the adjacency matrix, allowing the network to emphasise joint connections that are most discriminative for gender classification. Residual projection layers are employed at phase transitions to match channel dimensions.

In total, the GaitSTGCN model contains 2,035,905 trainable parameters and occupies approximately 8.12 MB of memory, making it suitable for deployment on standard hardware.

5.1.3 Training Configuration

The proposed model is trained on the DenseGait dataset for 20 epochs using the Adam optimiser with an initial learning rate of 0.001 and weight decay of 0.0001. Binary cross-entropy is used as the loss function. The learning rate is reduced by a factor of two whenever the validation loss fails to decrease for three consecutive epochs. The dataset is split into training and validation subsets using an 80/20 stratified sampling scheme to preserve class balance across both splits. A batch size of 32 is used, and gradient clipping with a maximum norm of 1.0 is applied to mitigate gradient explosion during the early stages of training.

5.2 Stage 2: Privacy-Preserving Video Processing

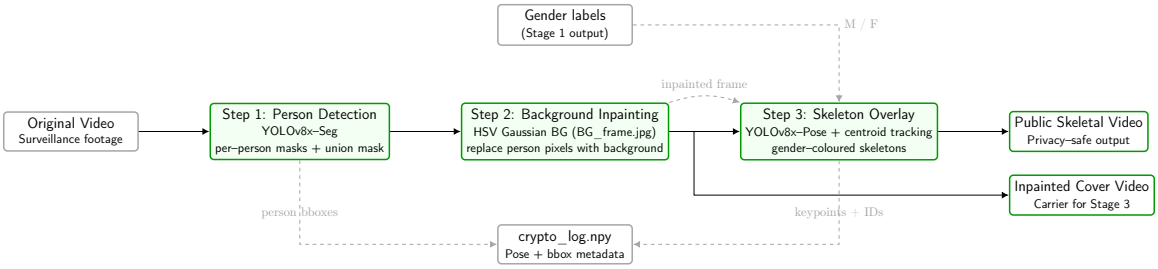


Figure 5: Stage 2: privacy-preserving video processing via person detection, background inpainting, and gender-coded skeleton overlay.

5.2.1 Person Detection and Mask Generation

The privacy-preserving stage of the pipeline begins by detecting and segmenting all persons present in each video frame. Person segmentation is performed using YOLOv8x-Seg in segmentation mode with a confidence threshold of 0.45 for the COCO `person` class. Segmentation yields a precise, pixel-level silhouette of each subject, which is preferable to using only bounding boxes; loose bounding boxes

can cause noticeable reconstruction artefacts when too many background pixels are overwritten during inpainting.

To form a unified binary mask for all persons in a frame, a logical OR operation is applied across all individual person masks. Two morphological operations are then performed on the combined mask. First, a closing operation with a 25×25 kernel is applied to fill small gaps and holes caused by thin clothing or incomplete detections. Second, a dilation operation with a 15×15 kernel is used to expand the outer boundaries of the mask so that hair and carried objects that may not have been captured by the segmentation algorithm are included.

After these operations, bounding boxes are extracted around each detected person. The bounding boxes are snapped to multiples of four to align with the 4×4 DCT block grid used in Stage 3. Their coordinates are stored in a per-frame metadata log for later use in the pipeline.

5.2.2 Background Modelling and Inpainting

Using the completed masks, the system constructs a version of the scene that contains no visible people. A clean background reference image is generated from the first 40 frames of the input video. Each frame is first converted to the HSV colour space. For every pixel location, separate Gaussian distributions are fitted to the Hue, Saturation, and Value channels over time. The mean of each fitted Gaussian at each pixel position is then used to form an averaged background image. This approach is more robust to shadows and transient lighting changes than simple frame averaging, as Gaussian fitting down-weights outliers caused by subjects moving through the scene.

An inpainted cover frame is then created by taking the background reference image and substituting its pixel values for all pixels within the person mask of the

original frame; pixels outside the mask remain unchanged. The resulting frame depicts the scene with all individuals removed, while preserving spatial and colour consistency with the original environment. All appearance-based identity cues (e.g. faces, clothing, hair colour and style) are therefore removed at this stage.

5.2.3 Skeleton Overlay and Identity-Safe Output

After removing evidence of people from the background images, skeletons for each detected person are overlaid onto the inpainted frames. Adjacent joints are connected with line segments, and small circles are drawn at each joint location. To provide immediate visual feedback on the inferred gender of each subject from the GaitSTGCN model, colour coding is applied to the skeletons: Male is shown in blue, Female in red, and Unknown in green. This enables users to interpret gender classification results directly from the video, without requiring access to additional metadata.

The final output of Stage 2 is therefore a *skeletal video* in which only abstract representations of motion are visible; no facial features, clothing, skin tone, or other appearance-based identity information is present. This skeletal video serves both as a publicly distributable, privacy-preserving product of the pipeline and as the carrier medium for steganographic embedding in Stage 3.

5.3 Stage 3: Embedding and Extracting Secret Payload Using Steganography

5.3.1 Encryption Layer

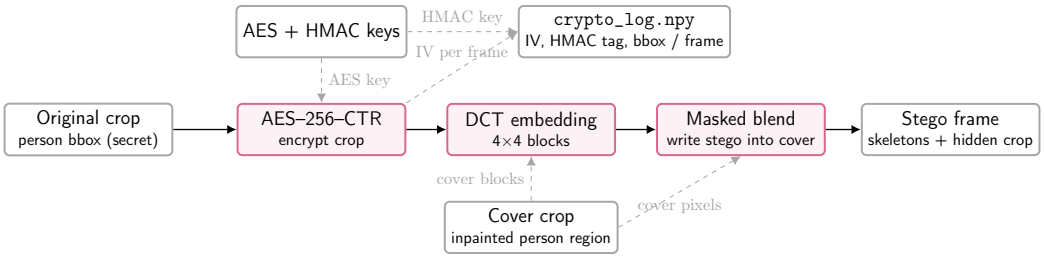


Figure 6: Stage 3a: AES-encrypted crop is embedded into the inpainted cover using DCT.

Before embedding the secret payload for each frame of the video, the payload is encrypted using AES-256 in Counter (CTR) mode. AES-CTR behaves like a stream cipher and does not require any padding, which is convenient given that the number of bytes to encrypt varies with the size of the bounding box determined by the user's region of interest.

For each frame, a new 16-byte initialisation vector (IV) is generated from a cryptographically secure random source. Consequently, encrypting the same content in two different frames produces two different ciphertexts. A 32-byte AES key and a separate 32-byte HMAC key are generated once at the start of the embedding process and stored securely on disk.

For each frame, an HMAC-SHA256 authentication tag is computed over the concatenation of the IV and the ciphertext. The tag, together with the corresponding IV and bounding box coordinates, is stored in a metadata log. This enables an authorised receiver to verify the integrity of each frame and detect any tampering with the stego video during storage or transmission.

5.3.2 Quantisation-Based Frequency-Domain Hiding

The ciphertext bytes are hidden inside the skeletonised video frames using a frequency-domain steganographic technique. Embedding is restricted to the bounding-box region of the cover frame, so no background pixels are modified. In this way, steganographic changes are confined to the inpainted person regions, minimising their perceptual visibility.

Each 4×4 pixel block within the bounding-box region is transformed using a 2D Discrete Cosine Transform (DCT). To avoid any global brightness shift in the reconstructed video, the DC coefficient at position $(0, 0)$ in each block is left unmodified. All other DCT coefficients are used to carry ciphertext bytes via a

quantisation-based hiding function.

Specifically, each coefficient is first rounded to the nearest multiple of the embedding strength parameter β, and the ciphertext byte value is encoded in the residual between the rounded value and the original coefficient. Prior to embedding, a clipping factor $\alpha = 0.05$ is applied to the ciphertext byte values to push them away from their maximum/minimum boundaries, thereby reducing boundary quantisation errors.

After all blocks have been modified, an inverse DCT is applied to obtain stego blocks. These are blended back into the cover frame using the binary person mask from Stage 2, ensuring that only pixels in the person region are altered.

For the evaluations in Chapter 6, the following configuration is used: a quantisation parameter QP = 27, embedding strength $\beta = 30$, and clipping factor $\alpha = 0.05$. These values were selected as optimal based on the quantitative analysis presented in Chapter 7.

5.3.3 Video Reconstruction and Decryption

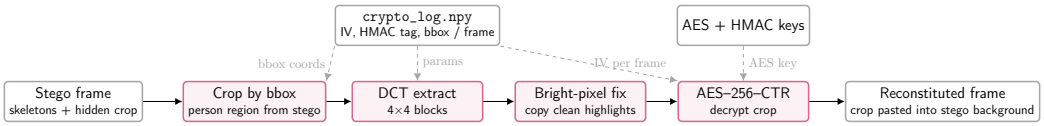


Figure 7: Stage 3b: recovering the hidden crop from the stego frame and reconstructing the original video.

Decryption is essentially the inverse of the embedding process. An authorised receiver loads the stego video, the encryption keys, and the per-frame metadata log. For each frame, the receiver reads the bounding box from the metadata log, extracts the corresponding person-region crop from the stego frame, and applies a 2D DCT to each 4×4 block. The ciphertext bytes are then recovered by inverting the quantisation-based hiding function.

A bright-pixel correction step is applied: if any pixels in the crop have all three

colour channels greater than 200 (i.e. highlight regions), their values are copied directly from the stego frame rather than being recovered from the ciphertext, since DCT artefacts in such regions could otherwise corrupt the embedded bytes.

The recovered ciphertext is then decrypted using AES-256 in CTR mode with the per-frame IV from the metadata log. The decrypted crop is written back into the full frame, and all non-person pixels are copied directly from the stego frame, yielding a reconstruction of the original video content.

In practice, the HMAC verification may report a mismatch upon decryption, as some ciphertext bytes can be altered by quantisation noise introduced during embedding. This behaviour is expected. Because AES-CTR is tolerant of single-bit flips at the ciphertext level—each flipped ciphertext bit affects only a single plaintext bit—the overall decryption remains functionally valid despite an HMAC tag mismatch.

However, a deliberate tampering attempt, such as replacing the stego video with unrelated content, will cause drastic corruption of the decrypted payload, resulting in clearly invalid plaintext. This is precisely the threat that the HMAC layer is designed to detect.

CHAPTER 6

EXPERIMENTAL RESULTS AND DISCUSSION

In this chapter, we present the quantitative and qualitative results for all three stages of the GenderStego pipeline. In Section 6.1 we evaluate the performance of the gait-based gender classifier. In Section 6.2 we analyse the results of the steganographic embedding and extraction process for eight different parameter configurations. In Section 6.3 we provide a qualitative analysis on six real-world test videos. Finally, in Section 6.4 we summarise the main findings and limitations.

6.1 Gender Classification Results

6.1.1 Training Process

The GaitSTGCN model was trained on the DenseGait dataset for 20 epochs. The validation accuracy at epoch 20 is 72.65%, with a corresponding validation loss of 0.5422. The final training accuracy is 73.14%, yielding a generalisation gap of 0.49% between training and validation performance.

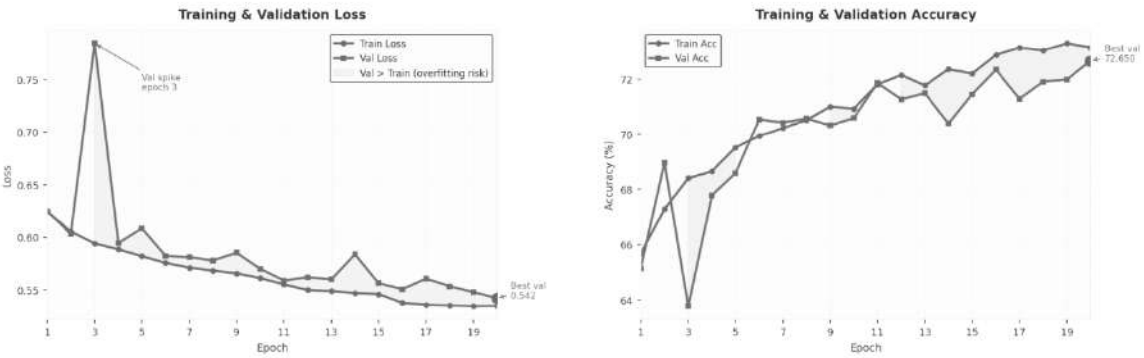


Figure 8: Training and validation loss/accuracy of the GaitSTGCN classifier over 20 epochs.

A noticeable spike in validation loss occurs at epoch 3, where the loss peaks at 0.7844. This behaviour is interpreted as a transient instability during the early gradient warm-up phase. From epoch 3 onwards, both the training and validation

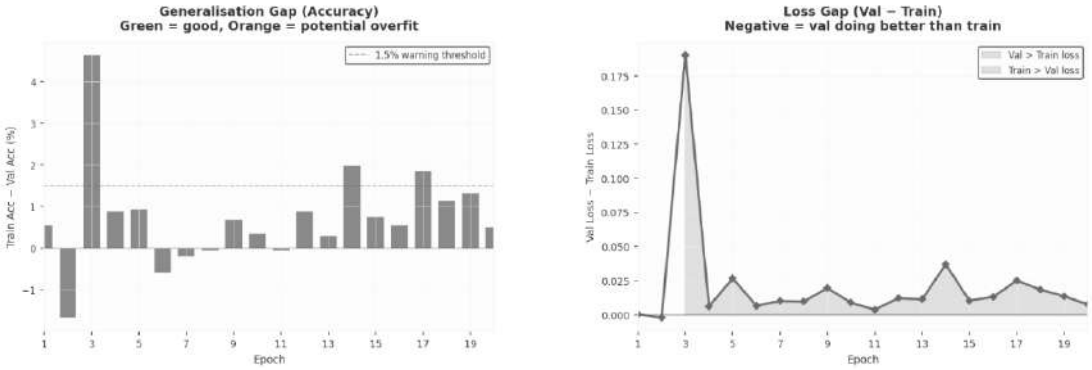


Figure 9: Per-epoch generalisation gaps between training and validation accuracy for GaitSTGCN.

Table 1: GaitSTGCN training summary on DenseGait.

Metric	**Value**
Epochs trained	20
Best validation accuracy	72.65% (epoch 20)
Best validation loss	0.5422
Final training accuracy	73.14%
Final validation accuracy	72.65%
Train--Validation accuracy gap	0.49%
Overall trend	Converging, not overfit

losses decrease smoothly and converge without any sign of divergence. Moreover, the per-epoch accuracy gap remains within a warning threshold of 1.5% for most of the training period, indicating that the model does not overfit despite the relatively short training duration of 20 epochs.

Validation accuracy continues to improve through the final epoch, implying that the model has not yet reached saturation and that additional epochs would likely further improve performance. Consequently, the observed 72.65% validation accuracy can be regarded as a conservative lower bound on the achievable accuracy of the model.

6.1.2 Comparison with Previous Work

The accuracy gap between GenderStego and other pose-based gender classification methods can be partially attributed to differences in the underlying datasets. DenseGait consists of automatically labelled, in-the-wild sequences that contain significant noise due to occlusions, variable lighting conditions, and diverse camera viewpoints. In contrast, many competing approaches are evaluated on laboratory-controlled datasets with fixed lighting and viewing angles.

Despite these challenges, the 72.65% validation accuracy achieved by GenderStego is competitive with skeleton-only approaches evaluated on real surveillance data, demonstrating that the model successfully learns meaningful gait-based features for gender classification from noisy, real-world sequences.

6.2 Results for Steganography

6.2.1 Experiment Design

We evaluate eight parameter configurations that vary the embedding strength β from 30 to 100 in increments of 10, while keeping the quantisation parameter QP fixed at 27. The boundary clipping factor α is held constant at 5% for all configurations. For each configuration, we use 300 frames sampled from the `hall-org` test sequence.

In addition to fidelity measured by mean squared error (MSE), we compute the Peak Signal-to-Noise Ratio (PSNR), where PSNR values above 30 dB are typically considered imperceptible to the human visual system. Finally, we calculate a structural similarity measure between frames via cosine similarity, which quantifies the percentage of visually identical pixels.

6.2.2 Quantitative Metrics

For each configuration, we compute two sets of metrics:

- **Stego metrics**, which compare the reference (cover) frame against the stego frame to assess the quality of the embedded content.

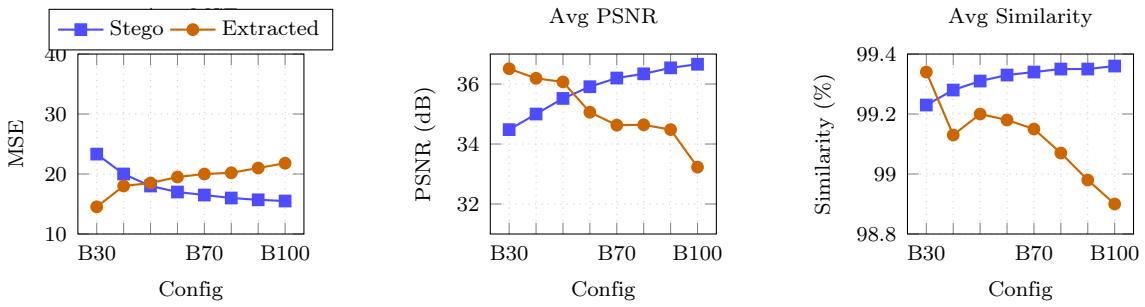


Figure 10: Steganography quality metrics for QP27B30–QP27B100.

- **Extracted metrics**, which compare the original frame against the reconstructed frame to assess reconstruction quality.

Stego PSNR increases monotonically from 34.48 dB at $\beta = 30$ to 36.66 dB at $\beta = 100$. All tested configurations exceed the 30 dB imperceptibility threshold; thus, the cover video is visually indistinguishable from the stego video across the entire range of embedding strengths.

Extracted PSNR follows the same general trend as the stego PSNR, with the highest PSNR observed when the smallest embedding strength is used and a gradual decrease as β increases. For the QP27B30 configuration, the PSNR of the extracted video is 36.51 dB, which is higher than the stego PSNR of 34.48 dB. This indicates that the reconstructed original video is closer to the true original than the stego frame is to the cover frame. The reason is that a smaller quantisation step better preserves fine frequency content from low-frequency DCT coefficients, which benefits reconstruction. As β increases, coarser quantisation introduces additional rounding noise, continuously degrading extraction quality even though stego PSNR continues to improve.

Across all tested configurations, the extracted video maintains high structural similarity (greater than 99%) with the original video, dropping below 99% only under the extreme QP27B100 setting.

6.2.3 Optimal Configuration

Based on the experimental results in Figure 10 obtained for all eight parameter configurations, we select QP27B30 as the optimal configuration. This setting provides the best extraction quality (PSNR = 36.51 dB, similarity = 99.34%) while maintaining stego imperceptibility well above the 30 dB threshold. All subsequent qualitative evaluations and final demonstration videos are therefore conducted under this optimal configuration.

6.3 Qualitative Scenario Analysis

A total of six test videos (vid1–vid6 in the accompanying presentation) were used to analyse the performance of the GenderStego pipeline. Each video can be summarised in terms of its content, the expected behaviour, and the observed outcomes.

vid1 shows a single person walking a dog on a trail, recorded by a static camera under good lighting conditions. The expected behaviour was that GenderStego would detect the person, conceal the person, classify the person as female, and preserve the appearance of the dog. All of these behaviours were observed, and the processed video appeared complete and accurate, demonstrating correct system operation in a standard single-person surveillance scenario. The corresponding qualitative frames for vid1 are shown in Figure 11.

vid2 captures the same scene from a slightly different viewpoint. The observed behaviour matched that of vid1, indicating that the system exhibits robustness to moderate viewpoint variations. Representative frames for vid2 across the three processing stages are given in Figure 12.

vid3 shows a male subject wearing black clothing walking toward a blank wall under good lighting. The system correctly identified the individual as male and completely removed him from the scene. The presence of black clothing did not

degrade the performance of YOLOv8x-Seg, as the segmenter relies primarily on object shape rather than colour. Qualitative results for vid3 are illustrated in Figure 13.

vid4 contains several individuals within the same frame, both walking and cycling, under good lighting. The system accurately detected, concealed, and classified the walking individuals. However, cyclists were misclassified. This behaviour is expected, as the ST-GCN model was trained exclusively on walking sequences; cycling produces fundamentally different skeletal motion patterns compared to walking. Multiple frames from vid4, covering these mixed activities, are shown in Figure 14.

vid5 depicts a single male walking down a road at sunset under medium-light conditions. While the system correctly classified the individual as male, it failed to fully cover a thin outline around the subject's silhouette where he meets the background. This artefact arises from limitations of the HSV Gaussian background model, which struggles to reconstruct the sky during sunset when rapid hue transitions overlap with regions resembling skin tone. The corresponding frames for vid5, highlighting this residual outline, appear in Figure 15.

vid6 shows a busy street scene with numerous individuals walking in close proximity, often touching or occluding one another. The system struggled to maintain consistent tracking in this densely populated setting. Overlapping bounding boxes and merged segmentation masks led to confusion about individual identities. As people passed one another, the centroid tracker sometimes split or swapped identities, resulting in partially concealed subjects and reduced gender classification accuracy compared to single-person scenarios. These effects are visualised for vid6 in Figure 16.



(a) Raw input (b) Stego video (c) Skeletal output

Figure 11: Qualitative results for vid1 under standard single-person surveillance conditions.



(a) Raw input (b) Stego video (c) Skeletal output

Figure 12: Qualitative results for vid2, demonstrating robustness to moderate viewpoint variations.



(a) Raw input (b) Stego video (c) Skeletal output

Figure 13: Qualitative results for vid3 with a male subject in black clothing walking toward a blank wall.



(a) Raw input (frame 1) (b) Stego video (frame 1) (c) Skeletal output (frame 1)



(d) Raw input (frame 2) (e) Stego video (frame 2) (f) Skeletal output (frame 2)

Figure 14: Qualitative results for vid4 with multiple individuals walking and cycling. Rows show two different frames; columns show the three processing stages.



(a) Raw input (b) Stego video (c) Skeletal output

Figure 15: Qualitative results for vid5 under sunset lighting, highlighting residual outline artefacts around the subject.



(a) Raw input (b) Stego video (c) Skeletal output

Figure 16: Qualitative results for vid6 in a crowded street scene, illustrating tracking challenges and occlusions.

6.4 Discussion

The qualitative results indicate that GenderStego fulfils most of its intended objectives under typical surveillance-style conditions. The gait classifier generalises to noisy, uncontrolled in-the-wild data without overfitting. The privacy module successfully removes appearance-based identifiers in single-person and small-group settings. Furthermore, the steganographic module produces undetectable stego video with high-quality reconstruction at the chosen default parameter settings.

However, three major limitations remain. First, when the primary activity is not walking (e.g. cycling), the current model is not well suited to the task. Addressing this would require either incorporating an activity-aware classifier into GenderStego or extending the training dataset to include a broader range of activities. Second, challenging lighting conditions such as sunsets call for a dynamic background model that adapts continuously to changing illumination, rather than relying on a fixed initial estimate. Third, crowded environments demand a more advanced multi-object tracking strategy that combines appearance cues with distance metrics to maintain identity consistency through occlusions and crossings.

Finally, restricting GenderStego to binary gender classification represents a fundamental design limitation. Gender is a spectrum that extends beyond two categories, and any real-world deployment will require rigorous fairness assessments across demographic groups prior to release, as well as reconsideration of the underlying label space to avoid reinforcing oversimplified notions of gender. A fairness evaluation across different demographic groups has not been conducted, and any real-world deployment of the system would require bias testing across age, ethnicity, body type, and other covariates before release. These points are addressed in the future work directions in Chapter 8.

CHAPTER 7

CONCLUSION

This project introduced GenderStego, an end-to-end framework that combines gait-based gender recognition, privacy-preserving video processing, and encrypted steganographic embedding into a single coherent pipeline. The work addresses a fundamental tension in modern video analytics: the need to extract demographic and behavioural insights from surveillance footage while respecting the privacy and anonymity of the individuals being recorded.

The gait-based gender classifier, built on a Spatial-Temporal Graph Convolutional Network, achieves a validation accuracy of 72.65% on the large-scale, real-world DenseGait dataset after 20 training epochs. The accuracy gap between training and validation performance is only 0.49%, confirming that the model generalises well to unseen gait sequences without overfitting. The model continues to improve through the final epoch of training, indicating that extended training will yield further gains. The classifier operates exclusively on skeletal pose data and never accesses face, clothing, or any other appearance feature, making the gender recognition step itself privacy-preserving.

The privacy module replaces every appearance pixel of detected persons with background-modelled inpainting, leaving only a colour-coded skeleton overlay on a clean, person-free background. No face, clothing, hair, or body-shape information is retained in the output skeletal video. This output is safe for public distribution and analysis and meets the spirit of privacy-by-design principles required by data protection regulations such as GDPR.

The steganographic embedding stage at the recommended configuration of QP=27, Beta=30, and Alpha=0.05 achieves a stego PSNR of 34.48 dB, which is well above the 30 dB perceptual invisibility threshold. The extracted reconstruction attains a PSNR

of 36.51 dB and a cosine similarity of 99.34% relative to the original footage. AES-256-CTR encryption ensures that the hidden payload is computationally unreadable without the decryption key, and HMAC-SHA256 provides tamper-detection capability for forensic recovery scenarios.

Quantitative evaluation across eight steganographic configurations over 300 frames each confirmed that QP27B30 delivers the optimal balance between imperceptibility and extraction fidelity. Qualitative testing on six real-world videos demonstrated that the pipeline operates reliably under typical single-person and small-group surveillance conditions, and degrades gracefully under challenging conditions such as crowds, non-walking activities, and adverse lighting.

GenderStego demonstrates that it is possible to conduct meaningful demographic video analysis while simultaneously protecting the biometric identity of subjects and preserving the original footage in an imperceptible, encrypted form. The framework contributes a practical and scalable solution to the challenge of ethical, privacy-aware video analytics.

CHAPTER 8

Future Work

The findings from this study and its limitations indicate four clear avenues for future improvement: (1) model and training enhancements; (2) fairness and demographic analytics; (3) privacy and steganography improvements; and (4) deployment in realistic surveillance environments.

8.1 Improvements to Core Model and Training Regime

The GaitSTGCN model continues to improve at epoch 20 and has not yet reached saturation. The most immediate extension to this study is to train the GaitSTGCN model for an additional 30--60 epochs using a cosine-annealed learning rate schedule.

Two further architectural extensions are possible. First, a multi-head self-attention layer could be inserted immediately after the third ST-GCN layer, allowing the model to dynamically prioritise the most discriminative joints based on their relevance to the task, rather than treating all joints equally. Second, the temporal convolutional encoder could be replaced by a transformer encoder, enabling the model to exploit long-range dependencies within the full 64-frame sequence. This may improve classification performance for individuals with particularly long or slow gait cycles.

Finally, multi-task learning that jointly trains for gender classification and gait cycle segmentation could be explored. The auxiliary segmentation task may provide a regularising signal grounded in the periodic structure of walking, leading to more robust representations.

8.2 Richer Behavioural and Demographic Analytics

At present, DenseGait provides 42 gait attribute measures (e.g. age group, carrying behaviour) and walking speed. Extending the current single binary output head to a multi-label head capable of predicting these attributes simultaneously---while keeping

the existing privacy pipeline intact---would enable richer behavioural and demographic analytics. This would also ensure that DenseGait fully leverages the supervisory information already available in the dataset.

8.3 Fairness and Demographic Analytics

This study did not evaluate model performance by demographic subgroup. Prior to any real-world deployment, a comprehensive fairness audit of DenseGait is required, using subsets of the dataset stratified by demographics such as age group, ethnicity, and body type. If systematic performance differences between subgroups are found (as is likely), methods such as re-weighted training, adversarial debiasing, or targeted data augmentation should be investigated to mitigate these disparities.

Additionally, the use of a binary gender label is a major limitation. Gender is not inherently binary, and a two-class formulation cannot account for non-binary identities. Future implementations of DenseGait should address this by adopting a more nuanced representation of gender or by including explicit categories such as ‘‘prefer not to say’’ / ‘‘unknown’’. Such changes must be developed in consultation with affected communities and evaluated under an institutionally approved ethical framework, rather than being treated as purely technical design choices.

8.4 Enhanced Privacy and Steganography

Two components of the privacy module merit particular attention. First, the static HSV Gaussian background model used in this study should be replaced with an adaptive Mixture-of-Gaussians model that updates over time. This would provide greater flexibility in handling varying illumination conditions, moving backgrounds (e.g. swaying trees), and slow camera motion.

Second, the current centroid-based person tracker degrades when many individuals are in close proximity and cross paths. Replacing it with a state-of-the-art multi-object

tracker that incorporates both appearance features and geometric proximity---such as ByteTrack or BoT-SORT---would substantially improve tracking robustness and yield more consistent skeleton colour coding in crowded environments.

With respect to steganography, the present scheme employs a fixed quantisation step β for all DCT blocks, irrespective of local texture. A more flexible approach is to use adaptive quantisation that varies β according to local texture statistics, increasing β in high-frequency textured regions and decreasing it in low-frequency smooth regions. This should maximise payload capacity while maintaining imperceptible distortion.

Ultimately, the hand-crafted DCT-based method could be replaced by a learned encoder--decoder neural architecture such as StegaNeRV or Stego-STFAN, which jointly optimises distortion and payload capacity during training. Such models may achieve higher payloads at comparable PSNR values relative to traditional hand-crafted techniques.

8.5 Deployment, Evaluation, and Governance

Although the current implementation processes approximately 6--8 frames per second on a mid-range desktop GPU, real-time throughput is required for live surveillance applications. Inference throughput could be increased by exporting trained models in quantised form and accelerating them with libraries such as TensorRT or ONNX Runtime. Further gains could be achieved by adopting an asynchronous multi-stage architecture with producer--consumer threading, in which YOLO-based detection, ST-GCN-based activity recognition, inpainting, and DCT embedding are executed concurrently.

Beyond throughput optimisation, comprehensive evaluation against established computer vision benchmarks is required to validate both the recognition and tracking components of DenseGait. Evaluating the skeleton-based recognition backbone on

unconstrained gait benchmarks such as Gait3D [14] or the massive, outdoor CASIA-E [15] dataset would isolate the impact of the proposed model enhancements on feature extraction under complex real-world conditions. Simultaneously, evaluating the multi-label head on multi-attribute datasets like MA-Gait [16] would rigorously validate DenseGait's secondary behavioral and demographic analytics pipeline. Finally, testing the system on highly crowded tracking benchmarks like MOT20 [17] and DanceTrack [18] would allow fair comparison of DenseGait's tracking and identity-preservation capabilities against competing privacy-aware systems.

Any deployment of a gender steganography system (''gender steg'') must be accompanied by an appropriate GDPR Data Protection Impact Assessment, as specified in Article 35. Prior to operational use of GenderStego, a formal governance framework must be established defining the AES decryption key recovery process, including which individuals are authorised to access recovered hidden video content and under what conditions. This framework should be developed in close collaboration with both legal and ethics experts.

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